

GNR 638 - Assignment 2 Technical report

Yasasvi Naidu (23B1037)
MVV Harsha Vardhan (23B0975)
B Ritwik (23B0954)

March 16, 2026

1 Experimental Setup and Methodology

1.1 Dataset Configuration

All experiments were conducted using the Aerial Images Dataset (AID), a large-scale dataset designed specifically for aerial scene classification. The dataset comprises 30 distinct scene classes, capturing varying resolutions, lighting conditions, and geographic locations to ensure robust feature evaluation. To simulate standard transfer learning constraints, the dataset was partitioned into a strict training and validation split, preventing data leakage during evaluation.

1.2 Training Hyperparameters and Hardware

The experiments were executed using cloud-based GPU acceleration (see the accompanying *Computational Budget Report* for exact hardware and runtime metrics). During the training phases (Scenarios 1, 2, and 3), models were trained for a standard duration of 30 epochs to observe convergence stability. We utilized standard loss functions (Cross-Entropy Loss) and evaluated performance primarily using top-1 validation accuracy. For few-shot learning (Scenario 3), the training data fraction was artificially constrained to test data-hunger.

2 Architectural Overview

To contextualize the transfer learning results, it is imperative to understand the structural differences of the chosen models.

2.1 ResNet50: The Residual Baseline

ResNet50 represents the classical deep CNN architecture. It utilizes residual blocks with skip connections to solve the vanishing gradient problem, allowing for the training of significantly deeper networks. It possesses a moderate inductive bias, relying heavily on standard spatial convolutions.

2.2 EfficientNet-B0: The Scaled Workhorse

EfficientNet-B0 is built upon the principle of compound scaling, uniformly balancing network depth, width, and resolution. It heavily utilizes Mobile Inverted Bottleneck convolutions (MBConv) and Squeeze-and-Excitation optimization. This heavily constrained design imparts a massive inductive bias, regularizing the network and forcing it to learn highly efficient representations.

2.3 ConvNeXt-Tiny: The Modernized CNN

ConvNeXt-Tiny was designed to bridge the gap between traditional CNNs and modern Vision Transformers (ViTs). It adopts macro-designs from ViTs and micro-designs (like larger kernel sizes and GELU activations). Because it mimics Transformers, it possesses a significantly weaker inductive bias compared to EfficientNet.

3 Scenario 1: Linear Probing Transfer

To test raw feature alignment, we trained a linear classifier on completely frozen models. This scenario strictly evaluates how well the pre-trained backbone transfers to the new task without any adaptation of its internal weights.

3.1 ResNet50

ResNet50 provides a strong baseline, establishing well-separated clusters, though slight class bleeding is visible in the confusion matrix.

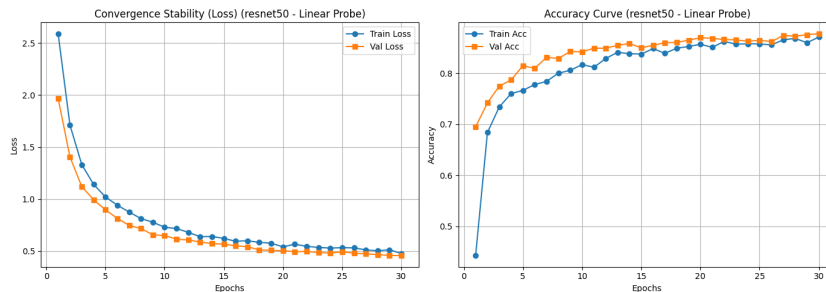
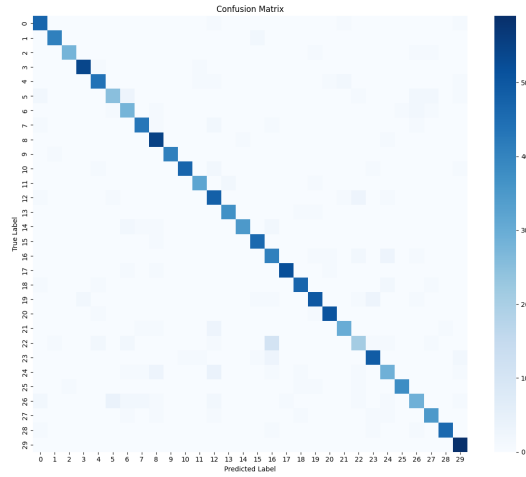
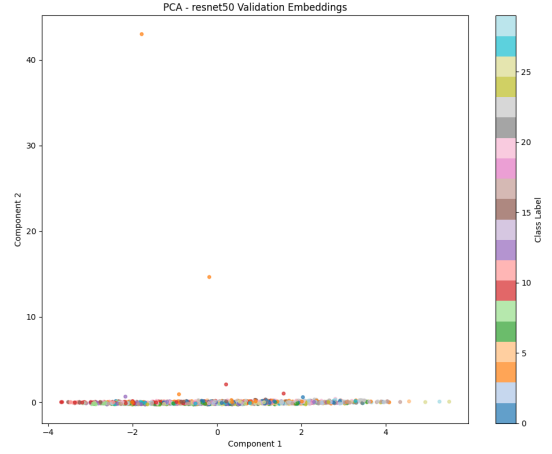


Figure 1: Scenario 1: ResNet50 Convergence



(a) Confusion Matrix



(b) PCA Embeddings

Figure 2: Scenario 1: ResNet50 Feature Analysis

3.2 EfficientNet-B0

EfficientNet-B0 exhibits the strongest baseline performance, displaying an incredibly clean confusion matrix diagonal and tight PCA clustering.

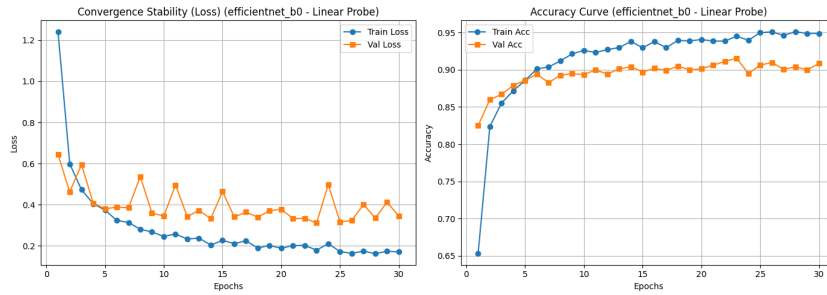
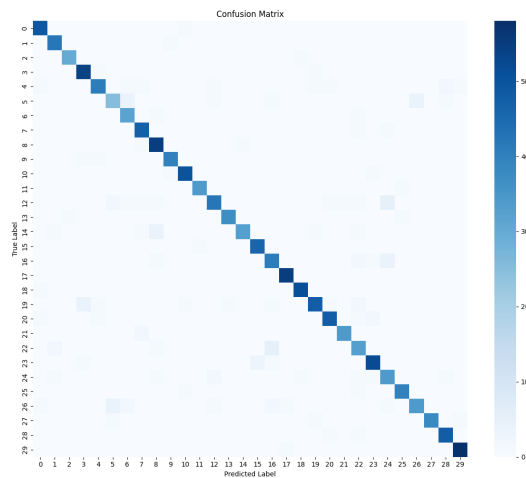
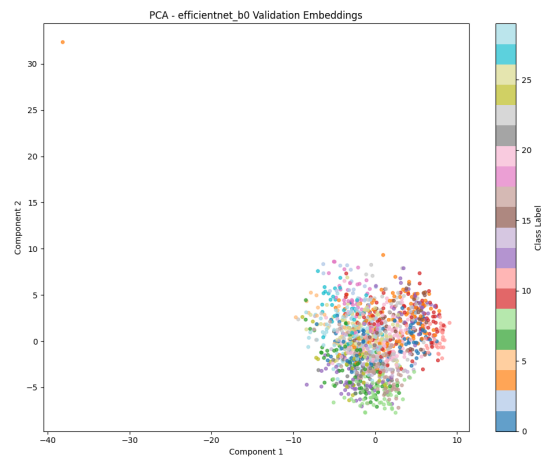


Figure 3: Scenario 1: EfficientNet-B0 Convergence



(a) Confusion Matrix



(b) PCA Embeddings

Figure 4: Scenario 1: EfficientNet-B0 Feature Analysis

3.3 ConvNeXt-Tiny

ConvNeXt-Tiny struggles as a frozen feature extractor. The overlapping PCA clusters and noisy confusion matrix indicate a significant domain gap.

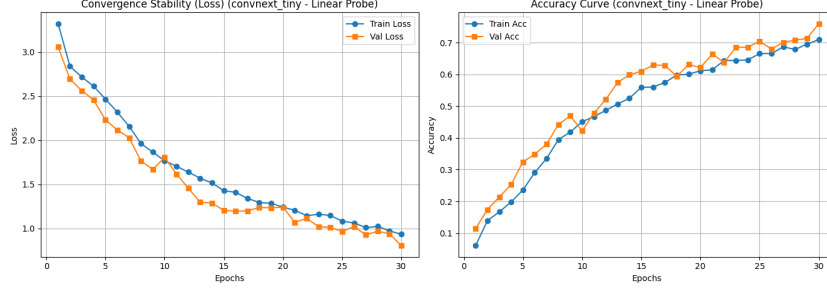
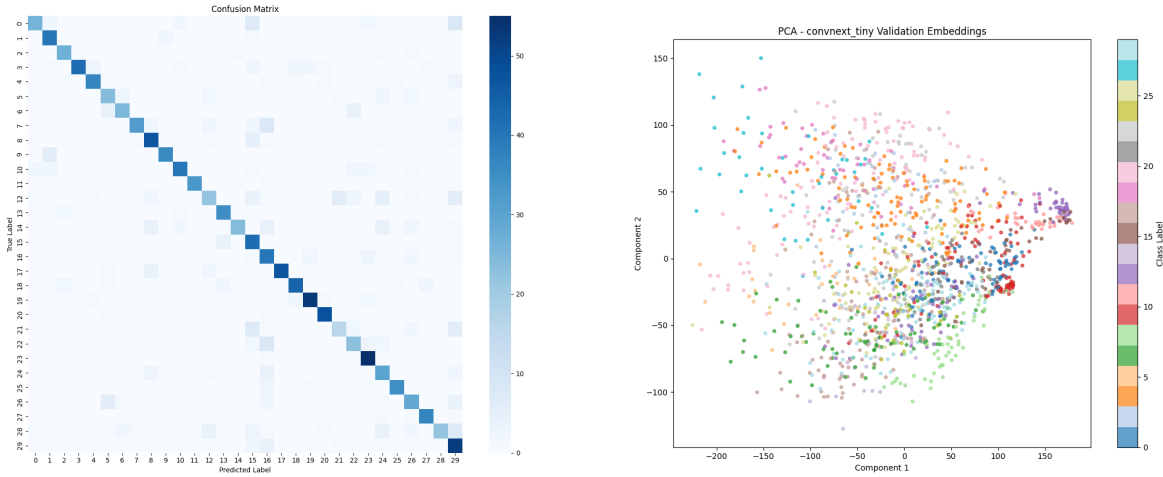


Figure 5: Scenario 1: ConvNeXt-Tiny Convergence



(a) Confusion Matrix

(b) PCA Embeddings

Figure 6: Scenario 1: ConvNeXt-Tiny Feature Analysis

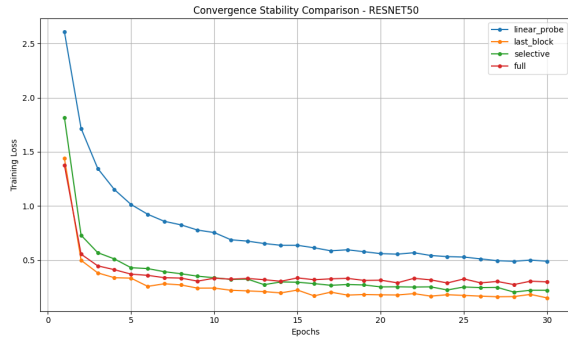
3.4 Analysis: Feature Separability and Transferability

In terms of feature separability, EfficientNet-B0 transfers exceptionally well without adaptation. Its PCA embeddings demonstrate high linear separability and distinct cluster compactness, proving its generalized ImageNet features map cleanly to aerial domains. Conversely, ConvNeXt-Tiny represents a distinct failure case under the frozen-backbone condition. Its PCA visualizations exhibit severe overlap and poor linear separability, indicating that its raw representations possess a significant domain gap and cannot effectively distinguish aerial macro-structures without adaptation.

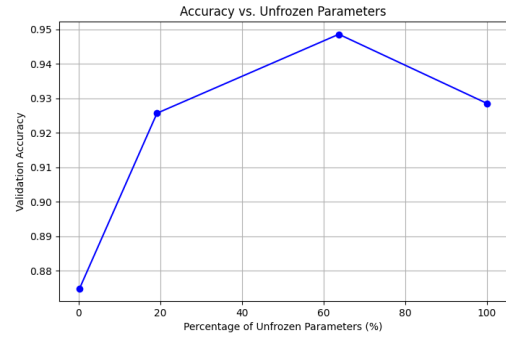
4 Scenario 2: Fine-Tuning Strategies

We evaluated how unfreezing different network depths (selective, last block, full) impacts performance.

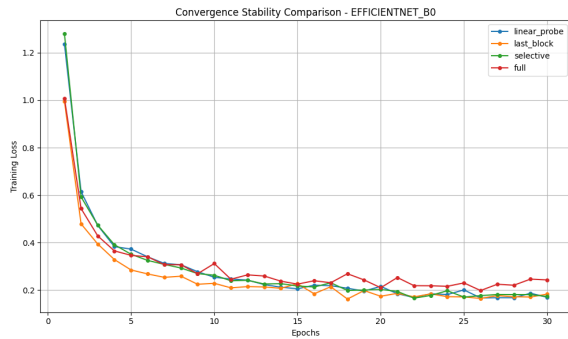
4.1 Convergence and Accuracy Tracking



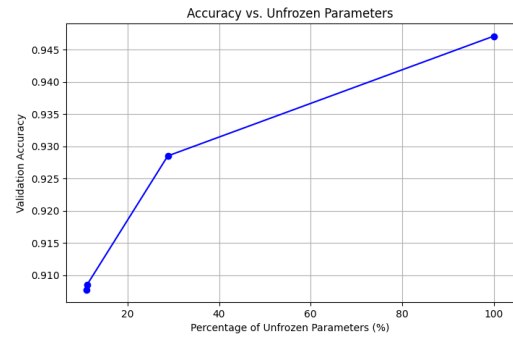
(a) ResNet50 Convergence



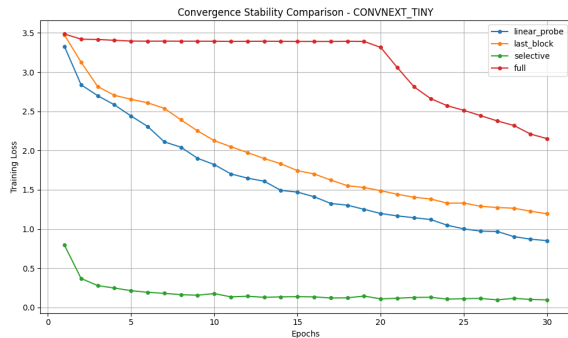
(b) ResNet50 Acc vs Params



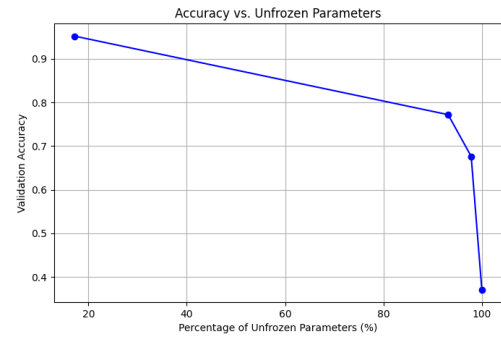
(c) EfficientNet-B0 Convergence



(d) EfficientNet-B0 Acc vs Params



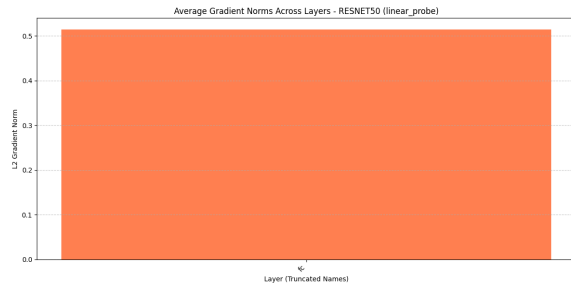
(e) ConvNeXt-Tiny Convergence



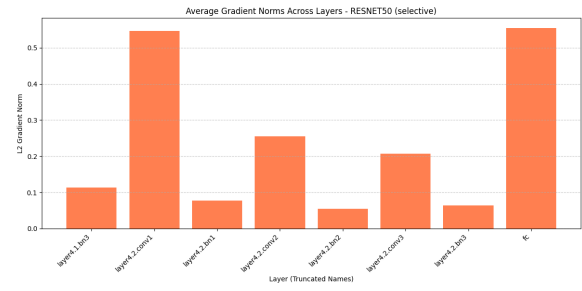
(f) ConvNeXt-Tiny Acc vs Params

Figure 7: Comparing convergence stability and unfrozen parameter efficiency across all models.

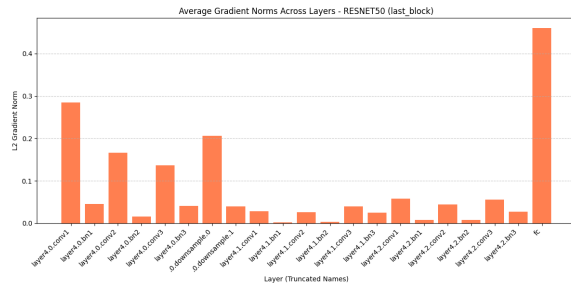
4.2 Gradient Norm Analysis



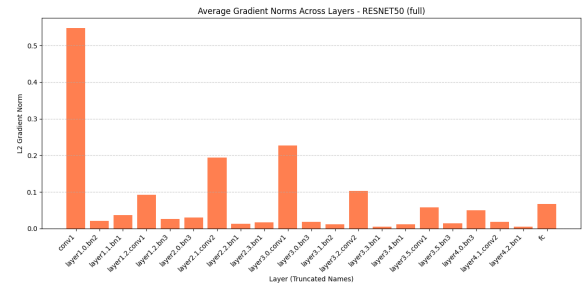
(a) ResNet50 (Linear Probe)



(b) ResNet50 (Selective)

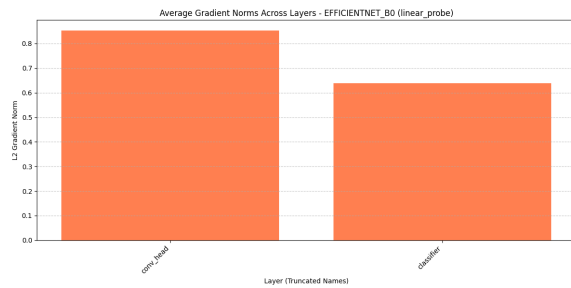


(c) ResNet50 (Last Block)

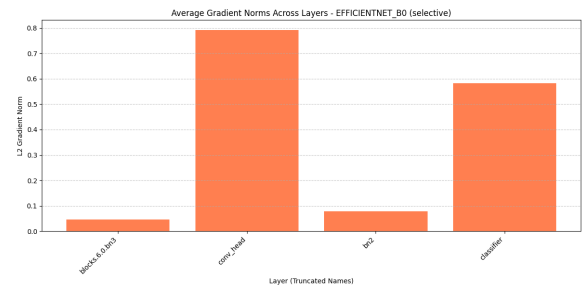


(d) ResNet50 (Full)

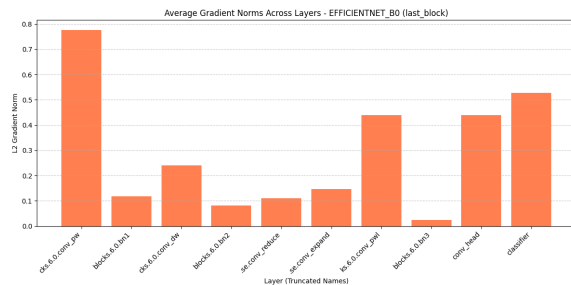
Figure 8: Gradient Norm Distributions across strategies for ResNet50



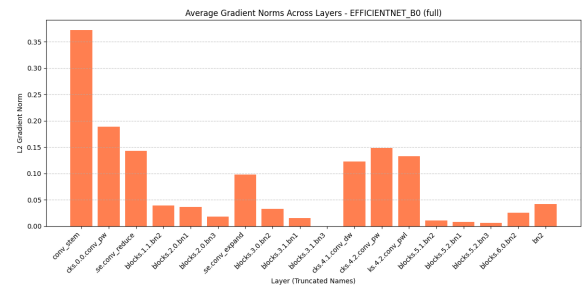
(a) EffNet-B0 (Linear Probe)



(b) EffNet-B0 (Selective)



(c) EffNet-B0 (Last Block)



(d) EffNet-B0 (Full)

Figure 9: Gradient Norm Distributions across strategies for EfficientNet-B0

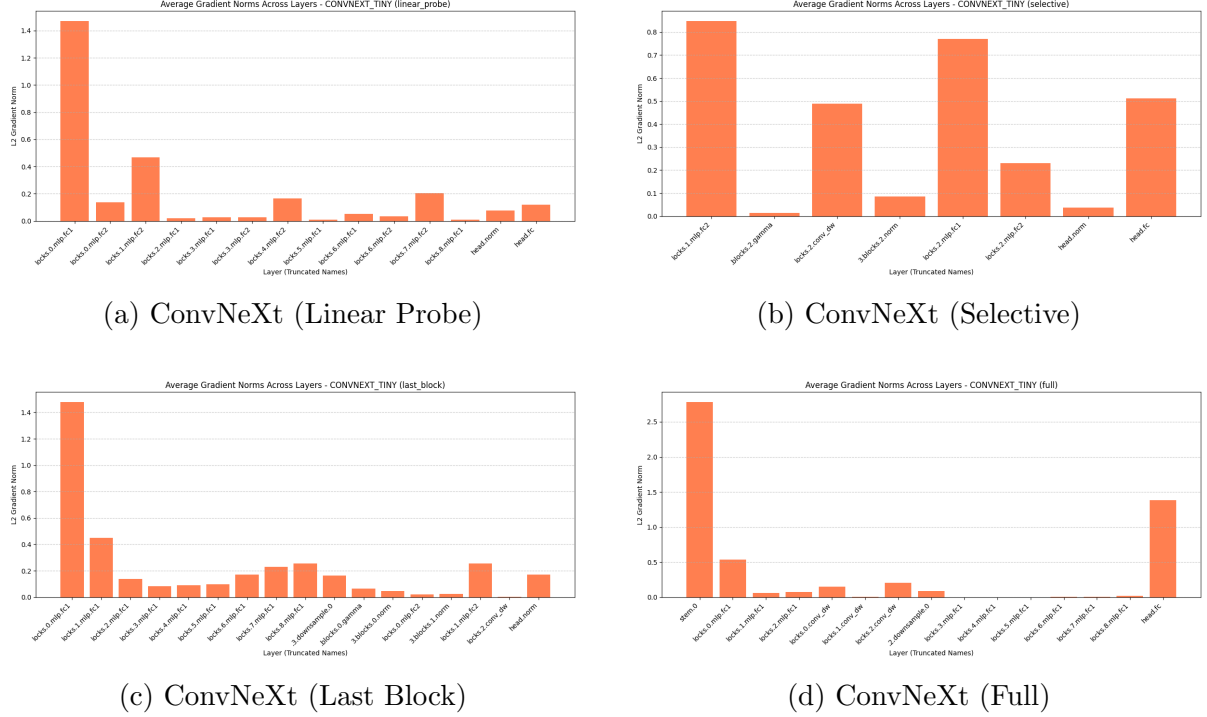


Figure 10: Gradient Norm Distributions across strategies for ConvNeXt-Tiny. Note the catastrophic spike in `stem.0` during full fine-tuning.

4.3 Analysis: Layer Adaptation and Catastrophic Forgetting

Modifying early layers versus deeper layers has profoundly different effects on representation quality, performance, and generalization. Deeper layers encode task-specific semantics; modifying them (as seen in the "Last Block" strategy) safely adapts the model to the aerial domain while preserving generalization. Conversely, modifying early layers risks destroying general-purpose visual primitives (edges, textures).

Our experiments identify that fine-tuning actively reduces model robustness and representation quality under the condition of full-network unfreezing without specialized optimization (e.g., learning rate warmups). The underlying mechanism for this failure is *catastrophic forgetting*. Because ConvNeXt-Tiny has a low inductive bias, applying standard learning rates to the entire network caused violent weight updates in early layers. This is evidenced by the massive gradient norm spike (> 2.5) in the `stem.0` layer during full fine-tuning, which shattered the model's foundational feature extractors and caused accuracy to plummet to $\sim 37\%$. ResNet50 requires a "Goldilocks" approach (adapting only the last block) to prevent similar, albeit milder, early-layer degradation.

5 Scenario 3: Few-Shot Learning Analysis

This scenario investigates how model capacity and architectural design influence performance in low-data settings. Table 1 documents the specific validation accuracy under each data regime, as well as the computed Relative Performance Drop (Δ).

Model	100% Data	20% Data	5% Data	Rel. Drop (Δ)
ResNet50	87.99%	78.84%	66.83%	24.05%
EfficientNet-B0	90.99%	82.84%	69.19%	23.96%
ConvNeXt-Tiny	74.27%	31.38%	22.59%	69.58%

Table 1: Validation accuracy and Relative Performance Drop (Δ) under data restriction.

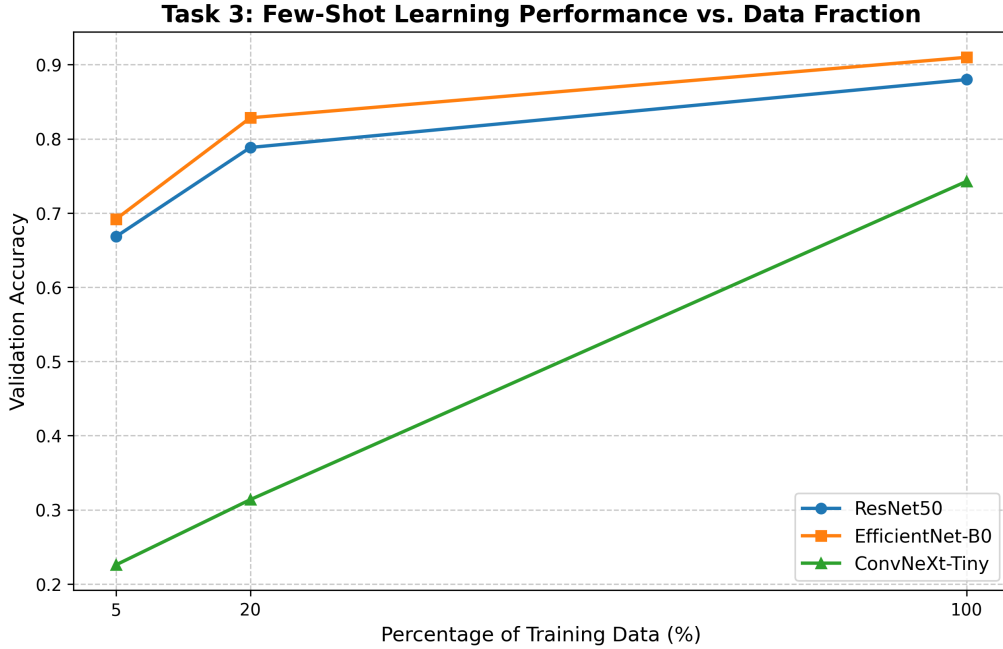


Figure 11: Few-Shot Learning Performance vs. Data Fraction.

5.1 Analysis: Model Capacity and Data Efficiency

The results decisively show that high parameter capacity does not guarantee data efficiency. EfficientNet-B0 demonstrates vastly stronger data efficiency than ConvNeXt-Tiny. Because EfficientNet relies on highly constrained architectural designs, it possesses a strong inductive bias that allows it to generalize rapidly from minimal examples. ConvNeXt-Tiny’s ViT-inspired architecture inherently lacks these spatial assumptions, resulting in a catastrophic $\sim 69.58\%$ relative performance drop when constrained to 5% of the data.

6 Scenario 4: Corruption Robustness Analysis

To quantify model resilience, we evaluate performance under noise, blur, and brightness corruptions. Table 2 displays the raw validation accuracy under each corruption condition. From these baseline metrics, we derived Table 3 (Corruption Error) and Table 4 (Relative Robustness) as required by our evaluation framework.

Model	Clean	Noise (0.05)	Noise (0.1)	Noise (0.2)	Blur	Brightness
ResNet50	87.21%	2.50%	2.50%	2.50%	39.31%	82.13%
EfficientNet-B0	90.21%	11.94%	3.00%	5.29%	33.74%	84.13%
ConvNeXt-Tiny	77.20%	31.74%	10.51%	4.43%	45.25%	63.47%

Table 2: Raw Validation Accuracy under Various Image Corruptions.

Model	Noise (s=0.05)	Noise (s=0.1)	Noise (s=0.2)	Motion Blur	Brightness
ResNet50	97.50%	97.50%	97.50%	60.69%	17.87%
EfficientNet-B0	88.06%	97.00%	94.71%	66.26%	15.87%
ConvNeXt-Tiny	68.26%	89.49%	95.57%	54.75%	36.53%

Table 3: Corruption Error ($1 - Acc_{corrupted}$) across simulated image corruptions.

Model	Noise (s=0.05)	Noise (s=0.1)	Noise (s=0.2)	Motion Blur	Brightness
ResNet50	2.87%	2.87%	2.87%	45.08%	94.18%
EfficientNet-B0	13.23%	3.33%	5.86%	37.40%	93.26%
ConvNeXt-Tiny	41.11%	13.61%	5.74%	58.61%	82.22%

Table 4: Relative Robustness ($Acc_{corrupted}/Acc_{clean}$) across simulated image corruptions.

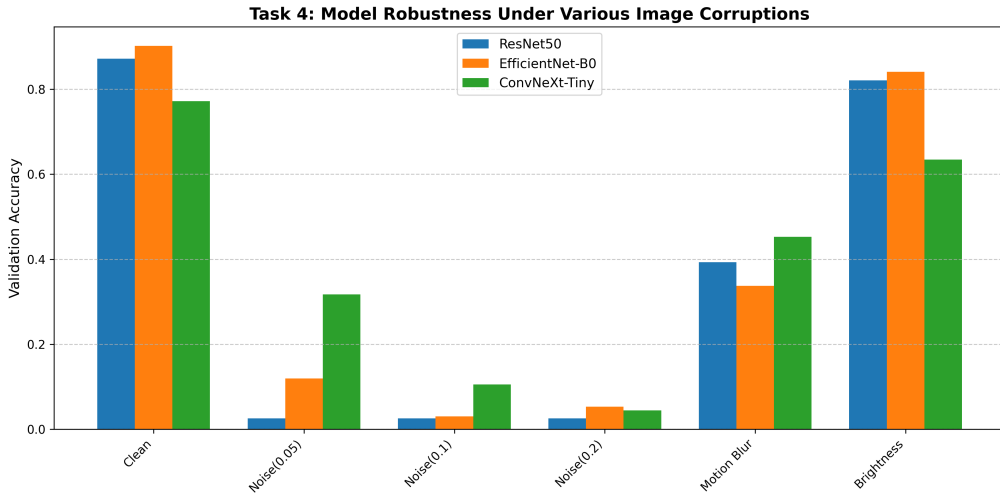


Figure 12: Model Robustness under Various Image Corruptions.

6.1 Analysis: Architectural Influence on Robustness

Comparing performance across models reveals distinct patterns regarding how architectural characteristics influence sensitivity to corruption. All three architectures exhibit severe vulnerability to high-frequency Gaussian noise. This is a known architectural flaw of standard CNNs, which over-rely on local, high-frequency textures rather than global shapes to make classifications. However, EfficientNet-B0 demonstrates the highest Relative Robustness against motion blur and brightness scaling, suggesting its Squeeze-and-Excitation modules help recalibrate channel features to maintain structural integrity under poor visibility.

7 Scenario 5: Layer-Wise Feature Probing

By training linear probes at intermediate layer depths, we map how semantics evolve through the networks. Table 5 documents the latent feature norm statistics extracted during evaluation.

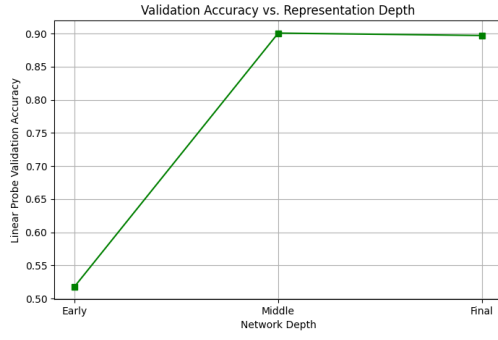
Model	Early (Mean $\pm$ Std)	Middle (Mean $\pm$ Std)	Final (Mean $\pm$ Std)
ResNet50	50.03 $\pm$ 1.37	49.25 $\pm$ 1.89	6.90 $\pm$ 0.94
EfficientNet-B0	3.99 $\pm$ 1.28	16.35 $\pm$ 2.26	29.03 $\pm$ 3.17
ConvNeXt-Tiny	11.25 $\pm$ 0.86	182.60 $\pm$ 16.27	10.51 $\pm$ 1.07

Table 5: Feature Norm Statistics across selected network depths.

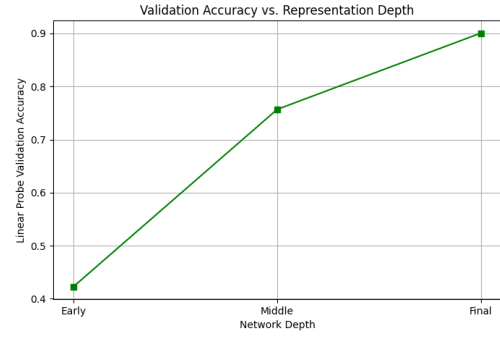
7.1 Layer Representation Accuracy

Model	Early Layer	Middle Layer	Final Layer
ResNet50	51.82%	90.06%	89.71%
EfficientNet-B0	42.32%	75.70%	90.06%
ConvNeXt-Tiny	79.06%	94.14%	91.85%

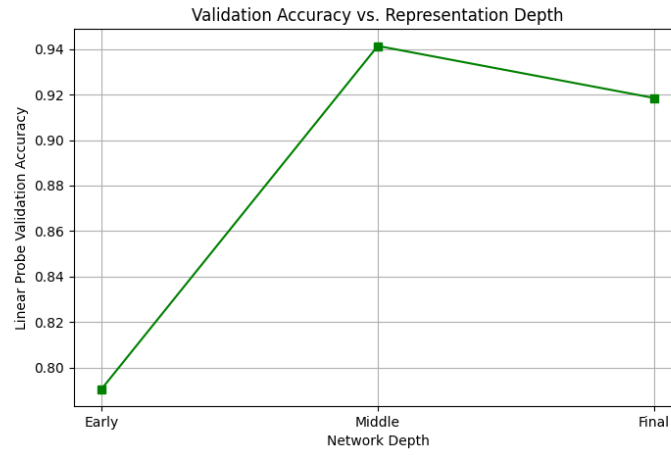
Table 6: Linear probe validation accuracy at different network depths.



(a) ResNet50 Layer Acc



(b) EfficientNet Layer Acc

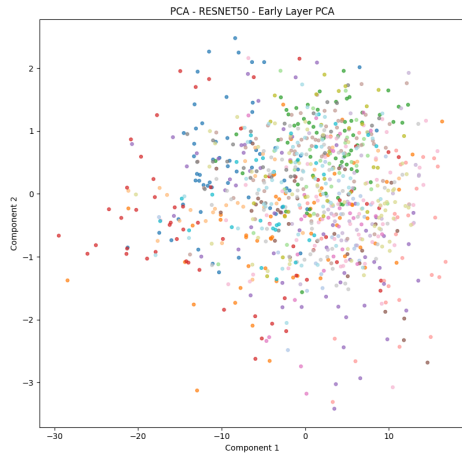


(c) ConvNeXt Layer Acc

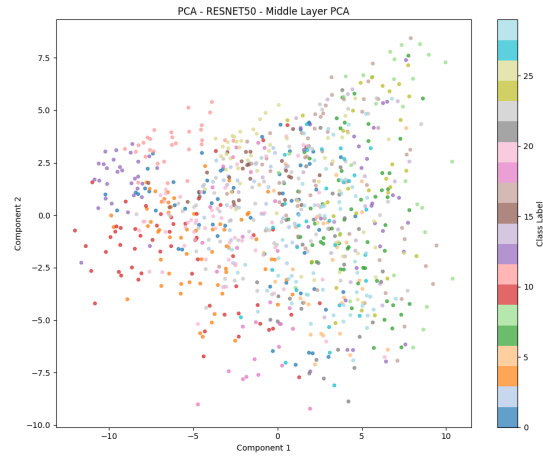
Figure 13: Accuracy as a function of representation depth.

7.2 Visualizing Semantic Evolution (PCA)

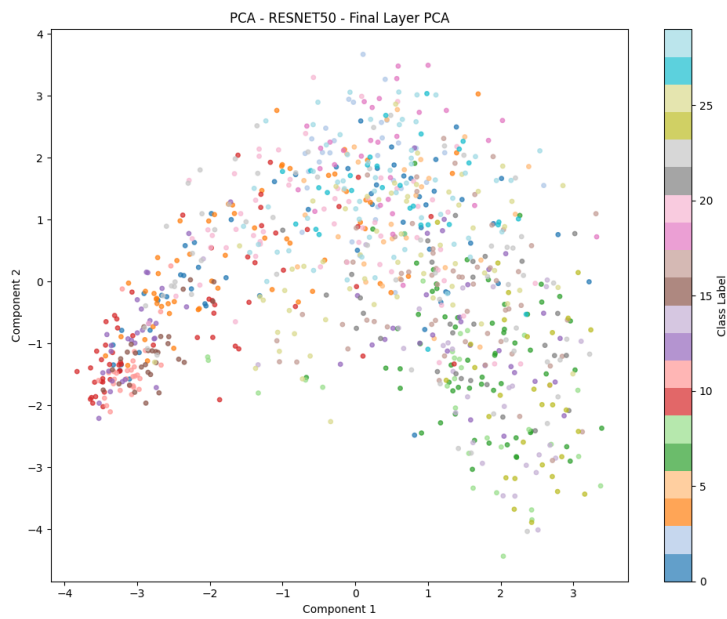
ResNet50 PCA Progression: Semantics peak in the middle layers and plateau.



(a) Early Layer



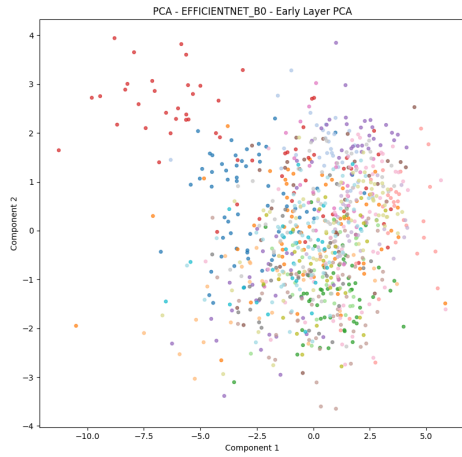
(b) Middle Layer



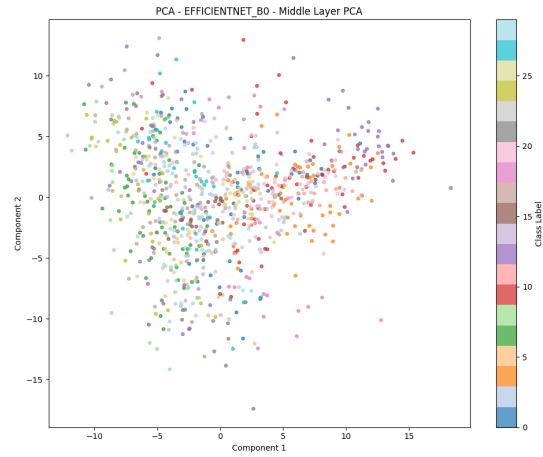
(c) Final Layer

Figure 14: ResNet50 Semantic Progression

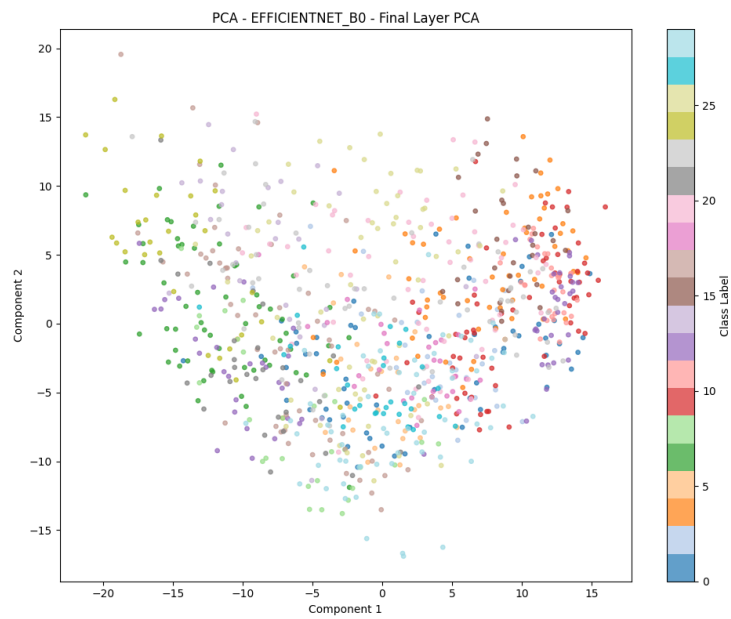
EfficientNet-B0 PCA Progression: Semantics cleanly separate all the way to the final layer.



(a) Early Layer



(b) Middle Layer



(c) Final Layer

Figure 15: EfficientNet-B0 Semantic Progression

ConvNeXt-Tiny PCA Progression: Semantics scramble at the end due to ImageNet hyper-specialization.

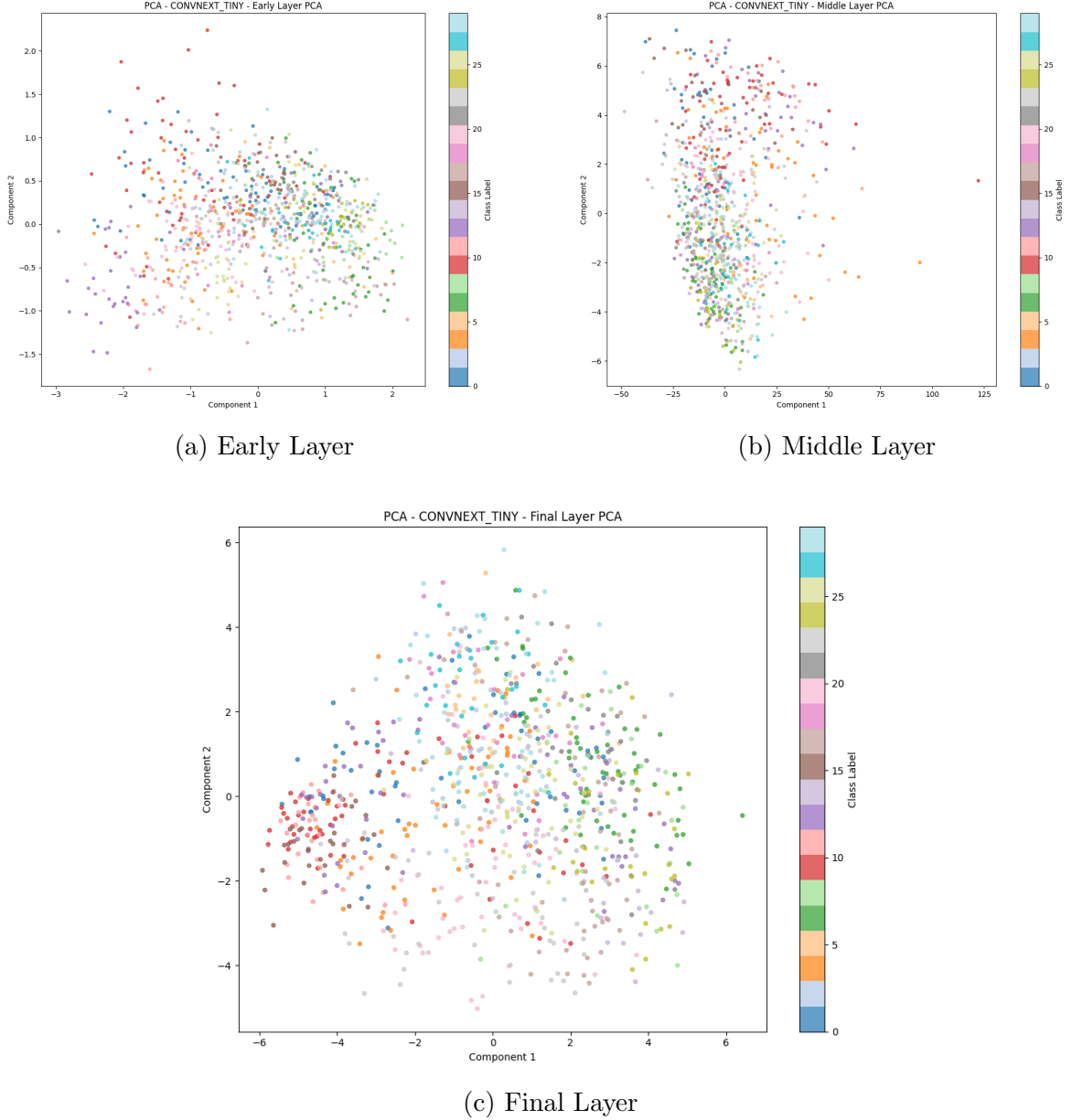


Figure 16: ConvNeXt-Tiny Semantic Progression

7.3 Analysis: Semantic Abstraction and Representation Hierarchy

Interpreting PCA visualizations alongside probing accuracy reveals that in early layers, all models exhibit high feature overlap and poor linear separability, as they are only extracting primitive edges. As depth increases to middle layers, we observe an increase in cluster compactness and semantic separation as primitives combine into recognizable aerial textures. This representation hierarchy directly relates to transferability. EfficientNet-B0 continuously increases cluster compactness through to the final layer, meaning its deepest hierarchy maintains generalized, highly robust features. Conversely, ConvNeXt-Tiny’s hierarchy breaks down at the end; its final representations scramble and lose cluster compactness due to hyper-specialization on original ImageNet classes, reducing both downstream transferability and robustness.